EART26201 | Principles of Geochemistry | Practical 3

Introduction to phase diagrams FEEDBACK

Task

This practical contains two questions about simple binary phase diagrams and the thermodynamics behind them. Each question should take about 60 minutes.

I strongly encourage you to discuss your work with each other as you go.

Demonstrators will be available throughout the practical class, and I will provide a short round-up at the end of the session. Feedback will be posted on Blackboard in due course.

Please share any comments you have as we go along, and I will do my best to adjust.

I suggest typing your answers into this document as well as annotating the associated blank phase diagrams.

Goals

At the end of this practical you should be able to:

- Label the solidus, liquidus and phase fields on a binary phase diagram
- Interpret simple binary phase diagrams with solid solution
- Use phase diagrams to explain equilibrium and fractional crystallisation and melting
- Use the lever rule to determine phase proportions on different phase diagrams
- Construct simple binary phase diagrams from $G-X$ diagrams

Other themes relevant to this week's content are reviewed in a Blackboard quiz.

You will revisit many of these themes in EART22200 (Igneous Minerals and Processes) and EART24200 (Metamorphic Minerals and Processes) later in the year.

Question 1: Interpreting a binary phase diagram with solid solution (60 minutes)

- You should have a copy of this plagioclase phase diagram on a separate sheet:

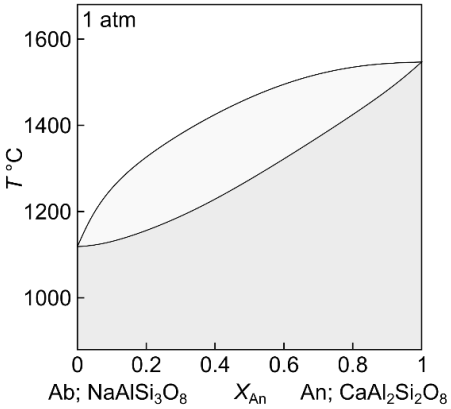
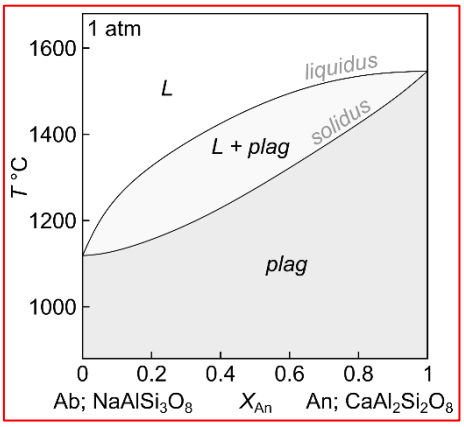


Figure 1: Plagioclase phase diagram

a) Label the liquidus and solidus.



b) Label the *L* (liquid), *L* + *plag* (plagioclase) and *plag* phase fields.

See answer to a)

c) How many components are there in the system, and what are they?

There are two components: albite (Ab) and anorthite (An)

d) How many degrees of freedom are there in a system with X_{An} = 0.5 at 1000 °C? How many at 1300 °C? You might find the Gibbs phase rule for an isobaric system useful:

$$f = c - \Phi + 1$$

At X_{An} = 0.5 and 1000 °C, $c = 2$, $\Phi = 1$ and therefore $f = 2$, which means that both *T* and X_{An} can be changed without changing the system (i.e. changing Φ)

At X_{An} = 0.5 and 1300 °C, $c = 2$, $\Phi = 2$ and therefore $f = 1$, which means that *T* and X_{An} cannot be changed without altering the system because although Φ will remain 2, the proportions of *L* and *plag* will change.

e) At what temperature does a liquid of composition X_{An} = 0.5 start to crystallise?

T ~ 1460 °C

f) What is the composition of the first plagioclase to crystallise from this X_{An} = 0.5 liquid?

X_{An} ~ 0.85

g) What is the composition of plagioclase in equilibrium with this liquid at 1400 °C?

X_{An} ~ 0.75

h) Calculate the equilibrium proportions of liquid and plagioclase in the system at 1400 °C. This requires the lever rule (see Lecture 3 | Part 3 for a recap) and a ruler.

~35% *plag*, ~65% *L*

i) What is the composition of the last plagioclase to form during equilibrium crystallisation of a liquid with an initial composition of X_{An} = 0.5?

X_{An} = 0.5

j) What is the composition of the liquid that forms this plagioclase?

X_{An} ~ 0.13

k) How would the last compositions of plagioclase and liquid differ if the system underwent fractional crystallisation instead of equilibrium crystallisation?

Both X_{An} = 0, i.e. Ab, instead of the values above

l) At what temperatures would a solid with a composition of X_{An} = 0.5 start to melt?

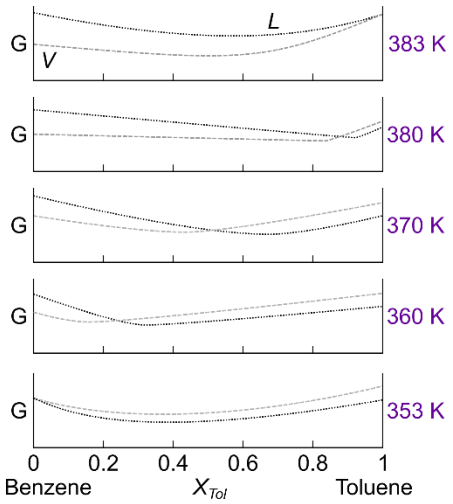
T ~ 1250 °C

m) What are the compositions of the last plagioclase crystals to melt during equilibrium and fractional melting?

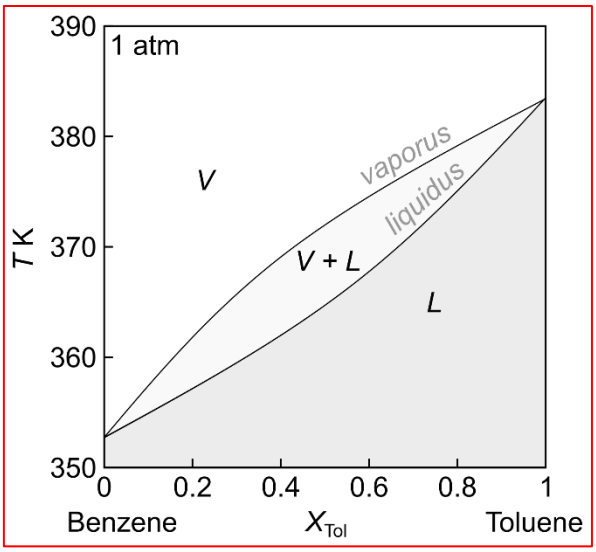
X_{An} ~ 0.85 (eqm) and X_{An} = 1 (frac)

Question 2: Constructing a phase diagram from $G-X$ diagrams (60 minutes)

- Although we are primarily concerned with high-T geochemical processes in my part of the course, remember that the thermodynamic principles we have been discussing are widely applicable. Benzene (Ben) and toluene (Tol) are organic solvents that mix when liquid but vaporise at different temperatures; this behaviour is not unlike that exhibited by olivine and plagioclase solid solutions.
- You should have a copy of these $G-X$ diagrams of the benzene-toluene system at 1 atm and five different temperatures (expressed in K) on a separate sheet where the curve for liquid is marked *L* and the curve for vapour is marked *V*:



a) Use the $G-X$ diagrams to construct a binary phase diagram (i.e., a $T-X$ diagram) for the benzene-toluene system at 1 atm on the sheet with an empty set of axes.



b) What is the equilibrium phase assemblage of system with the composition X_{Tol} = 0.4 at 365 K? What are equilibrium compositions and proportions of the phases present?

The equilibrium assemblage is *V* + *L*. *V* has a composition of X_{Tol} ~ 0.27 and *L* has a composition of X_{Tol} ~ 0.51. *V* and *L* are present in proportions of ~0.43 and ~0.57, respectively.

c) Quantitatively describe the equilibrium and fractional condensation of a vapour with the composition X_{Tol} = 0.4.

Equilibrium: First L condenses at ~369 K with an X_{Tol} of 0.63. All V has condensed by ~362 K when L has a composition of X_{Tol} = 0.4 and the last V has a composition of X_{Tol} ~ 0.2.

Fractional: First L condenses at ~369 K with an X_{Tol} of 0.63. L is continually removed from the system, with V and L compositions evolving along the vaporus and liquidus respectively until the system is pure Benzene (X_{Tol} = 0) at 353 K.

d) How could you synthesise pure toluene from a liquid with the composition X_{Tol} = 0.2?

Pure toluene could be produced by fractional vaporisation – heating the system and removing the vapour such that the final fraction of liquid – pure toluene – can be separated. This is fractional distillation.